

Pranay Katari

469-571-2396 | pranaykatari001@gmail.com | linkedin.com/in/pranaykatari | github.com/Pranay-Katari | [Portfolio](#)

EDUCATION

The University of Texas at Austin

Austin, TX

Bachelor of Science in Computer Science — Graduating May 2028

- Relevant Coursework: Software Engineering, Machine Learning, Elements of Statistics, Cloud Computing, Algorithms and Complexities, Operating Systems, Computer Architecture, Linear Algebra, Data Structures, Discrete Math

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, Java, C

Frameworks / AI: FastAPI, React, Next.js, Pandas, PyTorch, LangGraph, pgvector

Cloud / Data: AWS (EC2, Lambda, S3, DynamoDB, Secrets Manager), GCP Cloud Run, PostgreSQL, Supabase, Redis, Snowflake

Systems / Tools: Docker, Terraform, Git, GitHub, Bitbucket Pipelines, Linux, REST APIs, WebSockets

RELEVANT EXPERIENCE

Software Engineer Intern

June 2026 – August 2026

Invesco Ltd.

Houston, TX

- Enabled governed data dashboard access for **5,000+ EMEA users** by automating secure **AWS EC2 Gateway** provisioning with **Terraform** and **Bitbucket Pipelines** at a **\$2.47T AUM** global asset manager serving **120+ countries**.
- Built a serverless **Python AWS Lambda** integration using **AWS Secrets Manager**, **enterprise gateway APIs**, and **Snowflake** to securely route service requests to dynamic stored-procedure execution.
- Engineered and benchmarked **SQL**, **Python DataFrame**, **Snowpark API**, and **Snowflake OpenFlow** workflows, evaluating performance, scalability, and cost tradeoffs for production reporting systems.
- Used **Jira** and **Confluence** to manage development work, document technical processes, and coordinate delivery across engineering and analytics teams.

Software Developer

September 2025 – Present

Longhorn Developers

Austin, TX

- Built **React** features for **Degree Audit Plus**, a UT degree-planning platform with a **50,000+** userbase.
- Extended a **WXT/Vite Chrome Extension** with **TypeScript**, **Bun.js**, and **Tailwind CSS**.
- Utilized browser storage and **IndexedDB** to support degree audits, course search, dashboards, and planning workflows.
- Converted **Figma** designs into reusable components and partnered with Product Managers to ship student-facing features.

Software Engineer

May 2025 – September 2025

CodeSprout - codesprout.net

Austin, TX

- Integrated the **Piston API** with **Dockerized** sandboxes to enable secure execution of user-submitted code.
- Designed indexed **PostgreSQL** schemas to improve frequent-query performance and application responsiveness.
- Built **AI**-powered student features; earned **80%** positive teacher feedback during beta demonstrations.

PROJECTS

Aquarius Baskets | *FastAPI, Next.js, Supabase PostgreSQL/pgvector, Redis, LangGraph, Docker, Google Cloud Run* [Live Demo](#)

- Built a **FastAPI + NumPy/Pandas** backtester for **2–25 asset portfolios**, modeling fractional shares, rebalancing, costs, turnover, drawdown, and benchmark performance using Yahoo Finance and NYSE calendars.
- Optimized whole-percent weights across **500 Dirichlet/multinomial portfolios** with **1–50%** limits; used **3-fold expanding walk-forward validation** to measure unseen returns, drawdowns, and overfitting risk.
- Engineered an AI portfolio copilot with **LangGraph** and **pgvector-backed vector storage**, persisting user chats and retrieving relevant context for constrained basket proposals.
- Shipped a **Next.js + FastAPI** platform with Supabase OAuth, Redis caching, Dockerized Cloud Run, and per-user basket persistence for building, backtesting, and revisiting investment theses.

Real-Time BTC Prediction Market Terminal | *C++20, Python, FastAPI, Next.js, WebSockets, REST APIs* [GitHub](#)

- Engineered a persistent **C++20 streaming analytics engine** delivering **1-second** velocity, acceleration, rolling volatility, spread, target-margin, and adaptive-regime metrics for **Bitcoin 15-minute prediction markets**.
- Built a real-time **Python, FastAPI, WebSocket, and Next.js** pipeline unifying **4 live market-data feeds** into an interactive dashboard for cross-venue pricing, probability, and reversal-risk analysis.
- Developed and open-sourced a tested **Python trading CLI** supporting **YES/NO limit orders**, **IOC execution**, **position monitoring**, and **cancellations** with preview-first production safeguards.

Palladium Market Intelligence Agent | *Python, PyTorch, LangGraph, OpenAI API, Pandas, Tavily*

- Built a **Python/PyTorch** pipeline analyzing palladium returns, volume, rolling volatility, and volatility regimes through z-scores and a trained market-regime classifier.
- Developed a **multi-task LSTM** on normalized 60-day OHLCV sequences to predict price direction, five-day realized volatility, and candle-anomaly probability with chronological validation and early stopping.
- Designed a parallel **LangGraph multi-agent workflow** where supply, demand, and macro agents analyze market news before a judge agent produces attributed research using the **OpenAI API**.